

Activity

Case Analysis – Avoiding Misleading Proportions

Question 1: Truncated Y-Axis in Sales Growth

A retail company presents the following monthly sales (in ₹ lakhs):

Month Sales

January 95

February 97

March 99

April 101

May 103

June 105

The marketing team creates a bar chart with the **Y-axis starting at 90 instead of 0**. The chart visually suggests that June sales are almost twice those of January.

Tasks:

1. Explain why the chart is misleading.

Answer:

The Y-axis starts at 90 instead of 0, which exaggerates the differences between sales values. Although sales increased only from ₹95 lakhs to ₹105 lakhs (about 10.5%), the chart makes June sales appear almost twice as high as January.

2. Identify the visualization principle that has been violated.

Answer:

The principle of **honest and accurate representation of data** is violated. Bar charts should normally begin with a zero baseline to maintain proportional comparisons.

3. Recommend an improved visualization method.

Answer:

- Use a bar chart with the Y-axis starting at 0.
- Alternatively, use a line chart to show the monthly sales trend more accurately.

4. Describe how this misleading chart could influence business decisions.

Answer:

- Managers may believe sales have grown significantly.
- They may overestimate business performance.
- It could lead to unnecessary expansion or incorrect budgeting decisions.

Question 2: Pie Chart with Incorrect Proportions

A university reports student enrollment in four departments.

Department Students

Computer Science 420

Electronics 280

Mechanical 200

Civil 100

The designer accidentally creates a pie chart where the slices occupy approximately **50%, 30%, 15%, and 15%** instead of the correct proportions.

Tasks:

1. Calculate the correct percentage for each department.

Answer:

Total Students

$$420 + 280 + 200 + 100 = 1000$$

- Computer Science = **42%**
- Electronics = **28%**
- Mechanical = **20%**
- Civil = **10%**

2. Explain why the displayed chart is misleading.

Answer:

The chart shows slices of 50%, 30%, 15%, and 15%, which do not match the actual data. This gives a false impression of department sizes and misrepresents enrollment distribution.

3. Suggest an alternative chart that would improve comparison accuracy.

Answer:

A **bar chart** is a better choice because it allows easier comparison of student counts between departments.

4. Discuss how such errors could affect institutional planning.

Answer:

- Incorrect allocation of faculty.
- Improper classroom planning.
- Poor budgeting decisions.
- Misallocation of educational resources.

Question 3: Unequal Bar Widths in Revenue Comparison

A software company compares annual revenue.

Product Revenue (₹ Crores)

- A 60
- B 75
- C 90
- D 105

Instead of using equal-width bars, the designer creates bars with increasing widths, making Product D appear nearly **three times larger** than Product A.

Tasks:

1. Identify the graphical error.

Answer:

The chart uses **unequal bar widths**, making some bars appear larger than they actually are.

2. Explain how unequal bar widths distort perception.

Answer:

People perceive the area of the bar rather than just its height. Wider bars appear more important, exaggerating the revenue difference.

3. Redesign the visualization by specifying the correct design principles.

Answer:

- Use equal-width bars.
- Start the Y-axis at zero.
- Use a consistent scale.
- Label values clearly.
- Avoid unnecessary visual effects.

4. Discuss the possible consequences if investors rely on this chart.

Answer:

- Investors may overestimate Product D's performance.

- Poor investment decisions may occur.
- Company credibility may be damaged.
- Financial planning may become inaccurate.

Question 4: 3D Column Chart Distortion

An energy company reports renewable energy generation.

Source Energy (GWh)

Solar 450

Source Energy (GWh)

Wind 500

Hydro 550

Biomass 520

The analyst uses a **3D column chart** with perspective effects. Due to the viewing angle, the Hydro bar appears almost **40% larger** than Wind, although the actual difference is only **10%**.

Tasks:

1. Explain why 3D charts can create misleading proportions.

Answer:

Perspective and depth make some columns appear larger or smaller than their actual values, causing visual distortion.

2. Compare the perceived difference with the actual difference.

Answer:

Actual values:

- Wind = **500 GWh**
- Hydro = **550 GWh**

Actual difference = $550 - 500 = 50 \text{ GWh}$

Percentage difference = $(50 / 500) \times 100 = 10\%$

Perceived difference $\approx 40\%$

Therefore, the chart exaggerates the difference by about four times.

3. Recommend a better visualization.

Answer:

Use a **2D bar chart** or a **simple column chart** without 3D effects.

4. Discuss the importance of avoiding visual distortion in sustainability reporting.

Answer:

- Ensures transparency.
- Supports accurate policy decisions.
- Builds stakeholder trust.
- Enables fair comparison of renewable energy sources.

Question 5: Bubble Chart with Incorrect Scaling

A healthcare organization compares disease cases across five districts. **District Cases**

- A 150
- B 250
- C 350
- D 450

E 550

The analyst scales the **bubble radius directly** to the number of cases rather than scaling the **bubble area**. As a result, District E appears more than **ten times larger** than District A.

Tasks:

1. Explain why scaling the radius instead of the area is misleading.

Answer:

When the radius is scaled directly, the bubble area increases much faster because area is proportional to the square of the radius. This exaggerates differences between values.

2. Describe the correct method for proportional bubble sizing.

Answer:

Bubble **area** should be proportional to the data value.

Correct formula:

Radius $\propto$ Value

This ensures that the visual area accurately represents the data.

3. Suggest an alternative visualization suitable for this dataset.

Answer:

- Bar chart
- Dot plot
- Line chart (if showing trends)
- A **bar chart** is the most suitable for comparing disease cases across districts.

4. Discuss the potential impact of this visualization on public

health resource allocation.

Answer:

- Resources may be unfairly allocated.
- Some districts may receive excessive funding.
- Other districts may receive insufficient healthcare support.
- Public health planning may become ineffective.